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EMF

Electromagnetic field is the study of the effect of electric charges at rest or in motion.

Two kind of electric charges positive and negative.

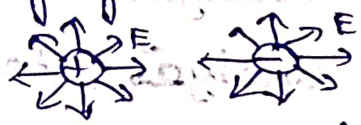
Both the positive and negative charges are the source of electric field.

When the charges are in motion the current is produced.

When the current is passed through a current carrying conductor the magnetic field is produced.

EMF is the combination of electric field and magnetic field.

1. (Electric field is perpendicular to magnetic field. $E \perp H$)

2. 

charge static condition.

The sources of electric field:

3. Sources of magnetic field: 

- Natural magnetic materials may be the charges are in moving condition.

4. Sources of electromagnetic field.

Natural sources in the sun

Human made sources x-Ray, Radio, mobile Phones etc

Basic of vectors:

②m or ④m

Scalar	Vector
The quantities which has only magnitude and no direction	The quantities which has magnitude and direction.
Eg: Mass of any object	Eg: Force, velocity, field intensity.
The scalar quantities are one dimensional quantities	Vector quantities are more than one dimensional quantities.
The scalar can be divided with another scalar quantities	Vector can be divided with another vector quantities.

Reviews of vector algebra:

Vector analysis is a mathematical tool used to analyze the electromagnetic fields.

Any physical quantity can be represented either as a scalar or a vector

Vector Algebras

1.) Vector addition

2.) Vector subtraction

3.) Vector multiplication → Dot product (scalar) ; → Cross product (vector)

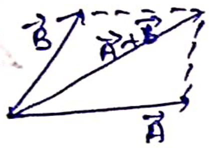
4) Differential vector operation

$\nabla V \rightarrow$ gradient

$\nabla \cdot \vec{A} \rightarrow$ Divergence

$\nabla \times \vec{A} \rightarrow$ Curl

Vector addition:



$$\vec{A} = A_x \vec{a_x} + A_y \vec{a_y} + A_z \vec{a_z}$$

$$\vec{B} = B_x \vec{a_x} + B_y \vec{a_y} + B_z \vec{a_z}$$

The sum of two or more vectors is the resultant vector

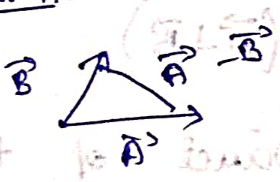
$$\vec{A} + \vec{B} = (A_x + B_x)\vec{a_x} + (A_y + B_y)\vec{a_y} + (A_z + B_z)\vec{a_z}$$

$$\vec{A} = 2\vec{a_x} + 3\vec{a_y} + 5\vec{a_z}$$

$$\vec{B} = \vec{a_x} + 2\vec{a_y} + 3\vec{a_z}$$

$$\vec{A} + \vec{B} = 3\vec{a_x} + 5\vec{a_y} + 8\vec{a_z}$$

Vector subtraction:



$$\vec{A} = 2\vec{a_x} + 3\vec{a_y} + 5\vec{a_z}$$

$$\vec{B} = \vec{a_x} + 2\vec{a_y} + 3\vec{a_z}$$

$$\vec{A} - \vec{B} = \vec{a_x} + \vec{a_y} + 2\vec{a_z}$$

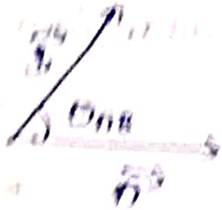
Vector multiplication:

i) Scalar or Dot product $(\vec{A}\cdot\vec{B})$

ii) Vector or cross product $(\vec{A}\times\vec{B})$

i) Scalar product (Dot)

$$\boxed{\vec{A}\cdot\vec{B} = AB\cos\theta_{AB}}$$



It is defined as the product of the two magnitudes and the cosine angle between them.

$$\vec{A} = A_x\vec{a_x} + A_y\vec{a_y} + A_z\vec{a_z}$$

$$\vec{B} = B_x\vec{a_x} + B_y\vec{a_y} + B_z\vec{a_z}$$

$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$$

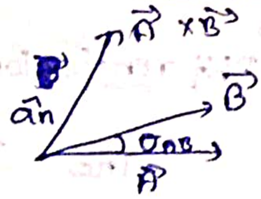
$$\left.\begin{array}{l} [\vec{a_x}\cdot\vec{a_x} = \vec{a_y}\cdot\vec{a_y} = \vec{a_z}\cdot\vec{a_z}] = 1 \\ [\vec{a_x}\cdot\vec{a_y} = \vec{a_y}\cdot\vec{a_z} = \vec{a_z}\cdot\vec{a_x}] = 0 \end{array}\right\} R$$

Conditions for two vectors to be perpendicular:

i) $\vec{A}\cdot\vec{B} = 0 \quad (\vec{A}\perp\vec{B})$

The dot product of two vectors is zero then the vectors are said to be perpendicular.

ii) Vector product: (cross)



$$\vec{A}\times\vec{B} = |AB\sin\theta_{AB}|\,\vec{a_n}$$

vector product of two vectors is a vector

Magnitude is the product of $\vec{A}$ and $\vec{B}$ and $\sin\theta_{AB}$.

Direction is perpendicular acting along the line of those two vectors.

$$\vec{A} = A_x\vec{a_x} + A_y\vec{a_y} + A_z\vec{a_z}$$
$$\vec{B} = B_x\vec{a_x} + B_y\vec{a_y} + B_z\vec{a_z}$$

$$\begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \vec{A}\times\vec{B}$$

<u>Property:</u>

i) It is not commutative $(\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A})$

ii) It is also not associative

<u>Conditions:</u>

$\vec{A} \parallel^{el} \vec{B}$

$\vec{A}\times\vec{B} = 0$ $\qquad\qquad$ $\hat{a_x}\times\hat{a_x} = 0$

$\Rightarrow\ \hat{a_x}\times\hat{a_y} = \hat{a_z}$

Unit vector $\quad \hat{a_y}\times\hat{a_z} = \hat{a_x}$ $\qquad$ Reverse

$x \quad y \quad z$ $\qquad\qquad$ $\hat{a_y}\times\hat{a_z} = -\hat{a_z}$

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Differentiation Operation:

The Differential vector operator is ∇

$$\nabla = \frac{\partial}{\partial x}\vec{a_x} + \frac{\partial}{\partial y}\vec{a_y} + \frac{\partial}{\partial z}\vec{a_z}$$

$\downarrow$
cartian differentiation

i) $\nabla V = \frac{\partial V}{\partial x}\vec{a_x} + \frac{\partial V}{\partial y}\vec{a_y} + \frac{\partial V}{\partial z}\vec{a_z}$

ii) $\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$

iii) $\nabla \times \vec{A} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$

Conditions for two vectors:

$\parallel^{el}$ & $\perp$ vector

$\vec{A}\cdot\vec{B} = 0$, $\vec{A} \perp \vec{B}$

$\vec{A}\times\vec{B} = 0$, $\vec{A} \parallel^{el} \vec{B}$

1. Show that the given vectors
$\vec{A} = 6\vec{a_x} + 2\vec{a_y} - 5\vec{a_z}$
$\vec{B} = 5\vec{a_x} - 5\vec{a_y} + 4\vec{a_z}$ are perpendicular to each other.

Soln:-

$\vec{A}\cdot\vec{B} = 0$ (perpendicular)

$\vec{A}\cdot\vec{B} = (6\vec{a_x} + 2\vec{a_y} - 5\vec{a_z})\cdot(5\vec{a_x} - 5\vec{a_y} + 4\vec{a_z})$

$\vec{A}\cdot\vec{B} = (6\vec{a_x}\cdot 5\vec{a_x}) + (2\vec{a_y}\cdot(-5\vec{a_y})) + (-5\vec{a_z}\cdot 4\vec{a_z})$

$\vec{A}\cdot\vec{B} = 30 - 10 - 20$

$\vec{A}\cdot\vec{B} = 0$. $\vec{A}$ & $\vec{B}$ vector are perpendicular

Hence showed

(2) Show the vector $\vec{A} = 4\vec{a_x} - 2\vec{a_y} + 2\vec{a_z}$, $\vec{B} = -6\vec{a_x} + 3\vec{a_y} - 3\vec{a_z}$ are parallel to each other.

Soln:-

$$(\parallel^{el}\ \vec{A} \times \vec{B}) = 0$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 4 & -2 & 2 \\ -6 & 3 & -3 \end{vmatrix}$$

$$= \vec{a_x}(6-6) - \vec{a_y}(-12+12) + \vec{a_z}(12-12)$$

$$\vec{A} \times \vec{B} = 0$$

$\vec{A}$ & $\vec{B}$ are parallel

∴ Hence showed.

Solenoidal & Irrotational Vector:

Solenoidal.

i) $\nabla \cdot \vec{A} = 0$ (Divergence is 0)

Irrotational: (curl free vector) (conservative vector)

$\nabla \times \vec{A} = 0$ (curl is 0)

(1) Show that the vector is solenoidal

$$\vec{H} = 3y^4 z^2 \vec{a_x} + 4x^3 z^2 \vec{a_y} + 3x^2 y^2 \vec{a_z}$$

Soln:-

solenoidal $(\nabla \cdot \vec{H} = 0)$

$$\nabla \cdot \vec{H} = \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z}$$

$$\nabla\cdot\vec{H} = \frac{\partial}{\partial x}\left[3y^4 z^2\right] + \frac{\partial}{\partial y}\left[4x^3 z^2\right] + \frac{\partial}{\partial z}\left[3x^2 y^2\right]$$

$$= 0 + 0 + 0$$

$$\nabla\cdot\vec{H} = 0$$

$\vec{H}$ is solenoidal.

∴ Hence showed

(2) Find the constant C such that the vector is solenoidal.

$\vec{A} = (x+ay)\vec{i} + (y+bz)\vec{j} + (x+cz)\vec{k}$ solenoidal.

Soln:-

$$\nabla\cdot\vec{A} = 0$$

$$\frac{\partial}{\partial x}\left[x+ay\right] + \frac{\partial}{\partial y}\left[y+bz\right] + \frac{\partial}{\partial z}\left[x+cz\right] = 0$$

$$1 + 1 + c = 0$$

$$\boxed{c = -2}$$

(3) Show that $\vec{H} = x^2\vec{a_x} + y^2\vec{a_y} + z^2\vec{a_z}$ is a conservative field.

Soln:-

Conservative $(\nabla\times\vec{H} = 0)$

$$\nabla\times\vec{H} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \vec{a_x}\left[\frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(y^2)\right] - \vec{a_y}\left[\frac{\partial}{\partial x}(z^2) - \frac{\partial}{\partial z}(x^2)\right]$$

$$+ \vec{a_z}\left[\frac{\partial}{\partial x}(y^2) - \frac{\partial}{\partial y}(x^2)\right]$$

$$= 0 - 0 + 0$$

$$\nabla \times \vec{H} = 0.$$

Hence showed

③ Verify whether the given vector $\vec{E} = yz\,\vec{a_x} + xz\,\vec{a_y} + xy\,\vec{a_z}$ is both solenoidal and irrotational vector.

Soln:

$$\nabla \cdot \vec{E} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy)$$

$$\nabla \cdot \vec{E} = 0 \quad [\text{solenoidal}]$$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xz & xy \end{vmatrix}$$

$$\nabla \times \vec{E} = 0$$

Irrotational

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Null Identities:

i) $\nabla \cdot \nabla \times \vec{H} = 0$

ii) $\nabla \times \nabla V = 0$

i) The divergence of curl of vector is 0.

ii) The curl of the gradient of any scalar is 0.

① Prove that $\nabla \cdot \nabla \times \vec{H} = 0$ (or divergence of curl of vector = 0)

$\vec{H}$ or any other variable

soln:-

$$\vec{H} = H_x \vec{a_x} + H_y \vec{a_y} + H_z \vec{a_z}$$

$$\nabla \times \vec{H} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ H_x & H_y & H_z \end{vmatrix}$$

$\vec{a_x}, \vec{a_y}, \vec{a_z}$ are unit vectors in particular direction

$$\nabla \times \vec{H} = \vec{a_x}\underbrace{\left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right]}_{A_x} - \vec{a_y}\underbrace{\left[\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z}\right]}_{A_y} + \vec{a_z}\underbrace{\left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right]}_{A_z}$$

$$\boxed{\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}}$$

$$\nabla \cdot \nabla \times \vec{H} = \nabla \cdot \vec{A}$$

$$= \frac{\partial}{\partial x}\left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right] - \frac{\partial}{\partial y}\left[\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z}\right] + \frac{\partial}{\partial z}\left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right]$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial H_z}{\partial y}\right) - \frac{\partial}{\partial x}\left(\frac{\partial H_y}{\partial z}\right) - \frac{\partial}{\partial y}\left(\frac{\partial H_z}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial H_x}{\partial z}\right) + \frac{\partial}{\partial z}\left(\frac{\partial H_y}{\partial x}\right) - \frac{\partial}{\partial z}\left(\frac{\partial H_x}{\partial y}\right)$$

$$= \cancel{\frac{\partial^2 H_z}{\partial x \partial y}} - \cancel{\frac{\partial^2 H_y}{\partial x \partial z}} - \cancel{\frac{\partial^2 H_z}{\partial x \partial y}} + \cancel{\frac{\partial^2 H_x}{\partial y \partial z}} + \cancel{\frac{\partial^2 H_y}{\partial x \partial z}} - \cancel{\frac{\partial^2 H_x}{\partial y \partial z}}$$

$$\nabla \cdot \nabla \times \vec{H} = 0$$

(2) Prove that $\nabla \times \nabla V = 0$

soln:-

Any scalar value the curl of gradient of Scalar is zero.

$$\nabla V = \frac{\partial V}{\partial x}\vec{a_x} + \frac{\partial V}{\partial y}\vec{a_y} + \frac{\partial V}{\partial z}\vec{a_z}$$

curl of ∇V

$$\nabla \times \nabla V = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}\left(\frac{\partial V}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial V}{\partial y}\right)\right]\vec{a_x}$$

$$- \left[\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial V}{\partial x}\right)\right]\vec{a_y} +$$

$$\left[\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial V}{\partial x}\right)\right]\vec{a_z}$$

$$= \left[\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial z \partial y}\right]\vec{a_x} - \left[\frac{\partial^2 V}{\partial x \partial z} - \frac{\partial^2 V}{\partial x \partial z}\right]\vec{a_y}$$

$$+ \left[\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial x \partial y}\right]\vec{a_z}$$

$$\nabla \times \nabla V = 0$$

Hence Null Identity is Proved

Co-ordinate System:

Three types of coordinate systems

↳ Cartesian or Rectangular coordinate system

↳ Cylindrical coordinate system

↳ spherical coordinate system.

Cartesian:

Represented by x, y, z

cylindrical:

Represented by ρ, ϕ, z

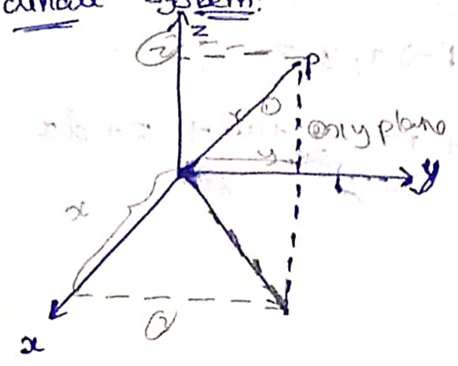
Spherical:

Represented by r, θ, ϕ

Co-ordinate system is a system which is used to represent a point in a space. It is used to describe a vector accurately with specific length, directions, angles, projections and components.

Any point is defined by three mutually perpendicular planes.

Rectangular coordinate System:



Draw a line from origin to Point P(which we assumed).

① Make a point 'P' in a space

② Draw a line straight away to xy plane.

③ Draw a line to the x and y.

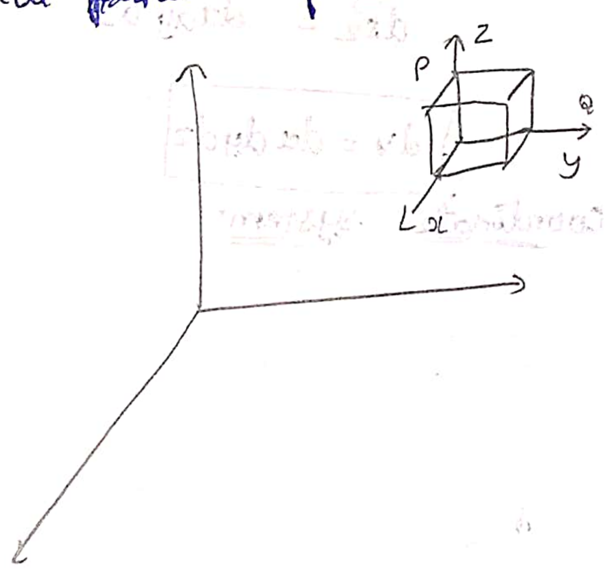
④ To get ~~...~~ z axis value

$$\vec{r} = x\,\vec{a_x} + y\,\vec{a_y} + z\,\vec{a_z}$$

x, y, z are the components of vectors

$\vec{a_x}, \vec{a_y}, \vec{a_z}$ are unit vectors

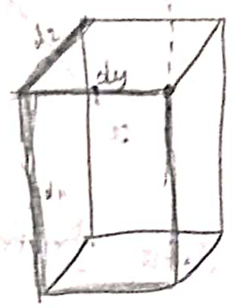
Differential ~~...~~ length surface and volume.



In case the length of point P, we use
(dx, dy, dz)

$P \to x, y, z$

$Q = x + dx,\ y + dy +\ z + dx$

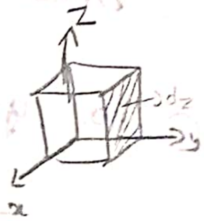


$dl = (x + dx,\ y + dy,\ z + dz) - (x, y, z)$

$\vec{dl} = dx\,\vec{a_x} + dy\,\vec{a_y} + dz\,\vec{a_z}$

The differential length is nothing by the diagonal value of Q.

$dl =$ end point - initial point



$|dl| = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$

$A = l \times b$

$\vec{ds_y} = dx\,dz\,\vec{a_y}$

$\vec{ds_x} = dy\,dz\,\vec{a_x}$

$\vec{ds_z} = dx\,dy\,\vec{a_z}$

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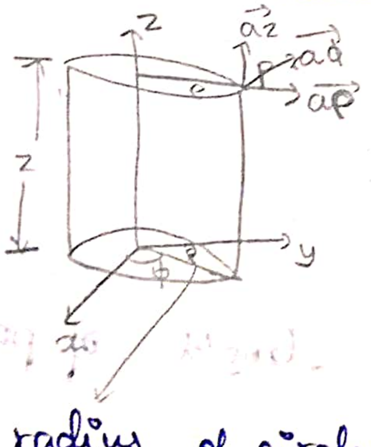
$\boxed{dv = dx\,dy\,dz}$

<u>Cylindrical</u> <u>coordinate</u> <u>system</u>:



ρ, ϕ, z

ϕ - angle made by ρ

ρ - radius of circle

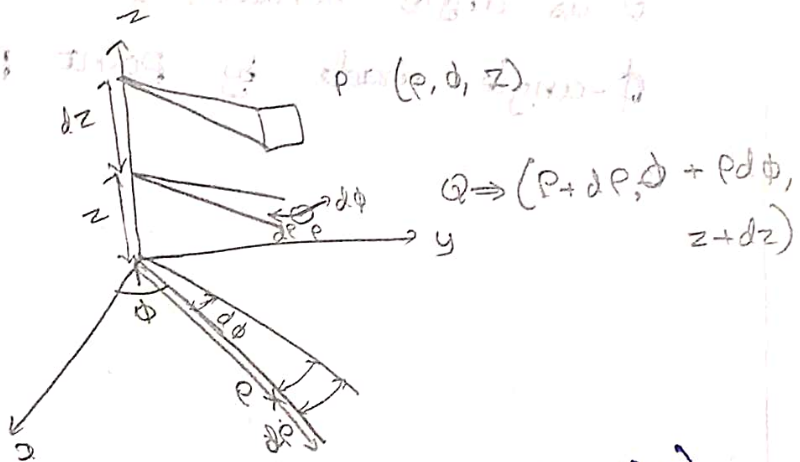
z - height of cylinder

In cylindrical coordinate system any point in space is considered as the intersection of three mutually perpendicular surfaces.

If ρ - constant, it will make circular cylinder surface.

If ϕ - constant, it will make a plane surface.

If z - constant, it will make another plane surface.

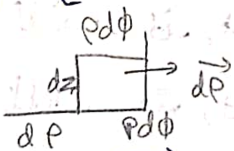


$$\vec{dl} = \left[(\rho + \partial\rho), \phi + \rho\partial\phi, (z + \partial z) \right] - [\rho, \phi, z]$$

$$\vec{dl} = d\rho \vec{a_\rho} + \rho d\phi \vec{a_\phi} + dz \vec{a_z}$$

$$|dl| = \sqrt{(d\rho)^2 + (\rho d\phi)^2 + (dz)^2}$$

Diff surface:



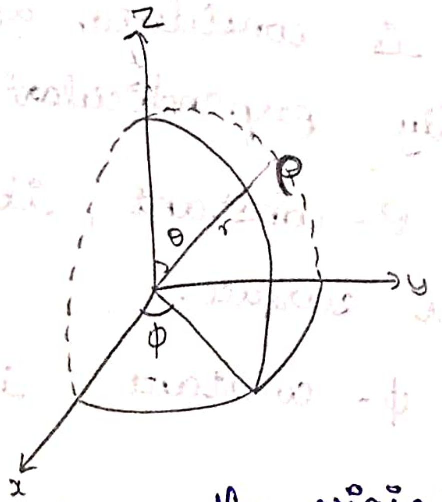
$$\vec{ds_\rho} = \rho d\phi dz \vec{d\rho}$$

$$\vec{ds_\phi} = d\rho dz \vec{d\phi}$$

$$\vec{ds_z} = \rho d\phi d\rho$$

$$dv = \rho d\rho d\phi dz$$

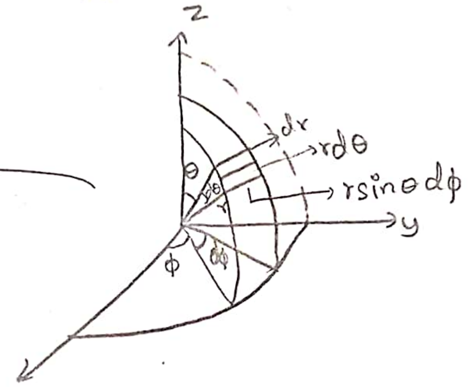
Spherical coordinate system:



r is the distance from the origin to point P.

θ is angle between z axis & radius r $(0 \le \theta \le \pi)$

ϕ-angle made by point P with respect to x-axis.



In spherical coordinate system any point P in space is considered as the intersection of three surfaces.

r is constant, it will create spherical surfaces

θ is constant, it will create conical surfaces,

ϕ is constant, it will create plane surfaces.

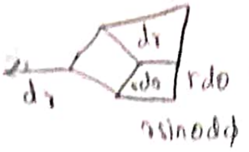
diagram here

Topic: Differential length

$$\vec{dl} = dr\,\vec{a_r} + r\,d\theta\,\vec{a_\theta} + r\sin\theta\,d\phi\,\vec{a_\phi}$$

$$|dl| = \sqrt{(dr)^2 + (rd\theta)^2 + (r\sin\theta\,d\phi)^2}$$

The. diff. surface



$$\vec{ds} = dr\,(r\,d\theta)\,\vec{a_\phi}$$

$$= r\,d\theta\,dr\,\vec{a_\phi}$$

$$\vec{ds} = (r\,d\theta)(r\sin\theta\,d\phi)\,\vec{a_r}$$

$$= r^2\sin\theta\,d\theta\,d\phi\,\vec{a_r}$$

$$\vec{ds} = r\sin\theta\,d\phi\,dr\,\vec{a_\theta}$$

$$dV = dr\,(r\,d\theta)\,(r\sin\theta\,d\phi)$$

$$dV = r^2\sin\theta\,dr\,d\theta\,d\phi$$

Conversion of coordinate systems: (related problems 4/8m)

i) Conversion of cartesian to cylindrical:

$x \quad \rho = \sqrt{x^2+y^2}$

$y \quad \phi = \tan^{-1}(y/x)$

$z \quad z = z$

ii) Conversion of Cartesian to spherical:

$x \quad r = \sqrt{x^2+y^2+z^2}$

$y \quad \theta = \cos^{-1}(z/r)$

$z \quad \phi = \tan^{-1}(y/x)$

iii) Conversion of cylindrical to Cartesian:

ρ $\qquad x = \rho \cos\phi$

ϕ $\qquad y = \rho \sin\phi$

z $\qquad z = z$

iv) Conversion of cylindrical to spherical:

ρ $\qquad \phi =$

ϕ $\qquad \phi =$

z $\qquad z =$

v) Conversion of spherical to Cartesian:

r $\qquad x = r\sin\theta \cos\phi$

θ $\qquad y = r\sin\theta \sin\phi$

ϕ $\qquad z = r\cos\theta$

1. Transform the cartesian coordinates 2, 1, 3 into cylindrical coordinates
2. Transform the given cylindrical coordinates $\rho = 1$, $\phi = 45°$, $z = 2$ into cartesian coordinates
3. ~~Transform~~ Express the given vector in cylindrical coordinate $\vec{A} = 4\vec{a_x} + 2\vec{a_y} + 3\vec{a_z}$.
4. Transform the cartesian coordinates to spherical coordinates $x = 2$, $y = 1$, $z = 3$
5. Find the cartesian coordinates $r = 4$, $\theta = 25°$, $\phi = 120°$.

1) soln:-

$x = 2$
$y = 1,$
$z = 3$

$\rho = \sqrt{x^2+y^2}$
$= \sqrt{4+1}$
$\rho = \sqrt{5}$

$\phi = \tan^{-1}(y/x) = \tan^{-1}(1/2) = 26.56°$

$z = z = 3$

∴ The cylindrical coordinats are $\sqrt{5}$, $26.56°$, 3

2) soln:-

$\rho = 1, \ \phi = 45°, \ z = 2$

$x = \rho \cos\phi$
$= 1 \cos 45°$
$x = 1/\sqrt{2}$

$y = \rho \sin\phi$
$y = 1 \sin 45°$
$y = 1/\sqrt{2}$

$z = z = 2$

∴ The cartesian coordinates are $\frac{1}{\sqrt{2}}, 1/\sqrt{2}, 2$

3) soln:-

$\vec{A} = 4\vec{a_x} + 2\vec{a_y} + 3\vec{a_z}$

$\rho = \sqrt{16+4}$
$\rho = \sqrt{20}$

$z = z = 3$

$\phi = \tan^{-1}(2/4)$
$= \tan^{-1}(1/2)$
$\phi = 26.56°$

4. soln:-

$$r = \sqrt{x^2+y^2+z^2}$$

$$\theta = \cos^{-1}(z/r)$$

$x=2, y=1, z=3$

$$\phi = \tan^{-1}(y/x)$$

$$r = \sqrt{4+1+9}$$

$$r = \sqrt{14}$$

$$\theta = \cos^{-1}(3/\sqrt{14})$$

$$\theta = 36.69^\circ$$

$$\phi = \tan^{-1}(y/x)$$

$$= \tan^{-1}(1/2)$$

$$\phi = 26.56^\circ$$

5. soln:-

$r=4$, $\theta = 25^\circ$, $\phi = 120^\circ$

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

$$x = 4(\sin(25^\circ))(\cos(120^\circ))$$

$$x = -0.845$$

$$y = 4(\sin(25^\circ))\sin(120^\circ)$$

$$y = 1.463$$

$$z = 4\cos(25^\circ)$$

$$z = 3.625$$

(1) Build the vector B in cartesian and Cylindrical systems. Given $B = \cos\theta \, \vec{a_\theta} + \vec{a_\phi}$, then find B at $(-3, 4, 0)$ and $(5, \pi/2, -2)$.

Soln:-

30/12/25.

Integrals of EMF:

Three types of integrals of field quantities over a region are used in emf.

i.) Line integral
ii.) surface integral
iii.) volume integral.

Line integral:

Line integral is mainly used to analyse an emf over a closed path or curved path.

surface integral:

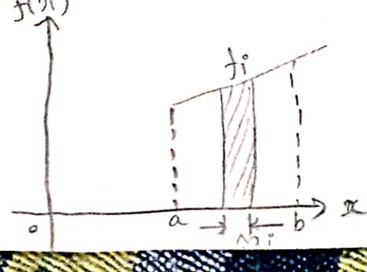
surface integral is mainly used to analyse an emf over a surface area.

Volume integral:

volume integral is mainly used to analyse an emf component over a volume.

i) Line integral:

It is the integral of the field quantity along a closed path.



$$\int_a^b f(x)\,dx = \lim_{\substack{n\to\infty \\ \Delta x_i \to 0}} \sum_{i=1}^{n} f_i \Delta x_i$$

Let $f(x)$ is a continuous function of x.

To define a line integral of $f(x)$ divide the integral from a to b into small segments of length $\Delta x \to 0$

Uses of Line Integral:

2m 1. To find a conservative field or Lamellar field.

$$\oint F\,dx = 0$$

Eg: static magnetic field

2. To calculate the workdone on a charge particle along a curve.

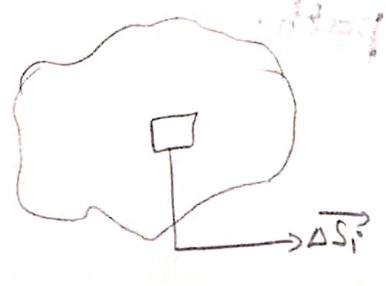
$$\oint E \cdot dl = V$$

Conservative field:

Line integral of field quantity around closed path is 0.

Line integral of electric field around closed path is equal to voltage.

ii) Surface Integral:

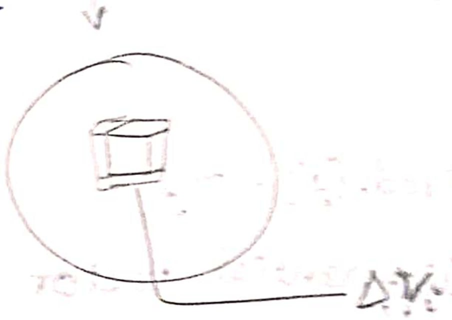


$$\int \vec{F} \cdot ds = \lim_{\substack{n \to \infty \\ \Delta s_i \to 0}} \sum_{i=1}^{n} \vec{F_i} \, \vec{\Delta s_i}$$

$$\iint F \cdot ds$$

To define the surface integral of $\vec{F}$ divide surface into small surface elements 'Δs'.

ii) Volume integral:



$$\int \vec{F} \cdot dv = \lim_{\substack{n \to \infty \\ \Delta V_i \to 0}} \sum_{i=1}^{n} \vec{F_i} \, \Delta V_i$$

$$\iiint \vec{F} \cdot dv$$

. It is the integration over a volume of field quantity.

Uses:

1. The fundamental laws of emf are expressed in integrals.

Eg: → Divergence theorem:

$$\iiint_V \nabla \cdot A \, dV = \iint_S A \cdot ds$$

• To relate volume integral to the surface integral.

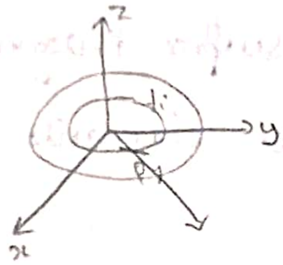
2. Eg: Stoke's theorem:

$$\oint H \cdot dl = \iint_S (\nabla \times H) \cdot ds$$

Relate line integral to surface integral.

Gradient of scalar: (cartesian / rectangular)

It represent the maximum rate of change of scalar field.



It is represented as gradient of f is equal to ∇f.

$$\text{Grad}(f) = \nabla f.$$

where $\nabla \rightarrow$ differential vector operator.

$$\nabla = \frac{\partial}{\partial x}\vec{a_x} + \frac{\partial}{\partial y}\vec{a_y} + \frac{\partial}{\partial z}\vec{a_z}$$

$$\nabla f = \frac{\partial f}{\partial x}\vec{a_x} + \frac{\partial f}{\partial y}\vec{a_y} + \frac{\partial f}{\partial z}\vec{a_z}$$

The gradient of scalar is a vector field. the magnitude is the rate of change of scalar quantity.

Gradient always points in the direction of the maximum rate of change of f (given scalar function).

Gradient in cylindrical coordinates:

$$\nabla f = \frac{\partial f}{\partial \rho}\vec{a_\rho} + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\vec{a_\phi} + \frac{\partial f}{\partial z}\vec{a_z}$$

Represented by (ρ, ϕ, z).

Gradient in spherical coordinates:

$$\nabla f = \frac{\partial f}{\partial r}\vec{a_r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\vec{a_\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\vec{a_\phi}$$

It is denoted as (r, θ, ϕ).

① Find the gradient of scalar system $f = x^2y + e^z$ $(1, 2, -1)$

soln:-

$$f = x^2y + e^z$$

$$\nabla f = \frac{\partial f}{\partial x}\vec{a_x} + \frac{\partial f}{\partial y}\vec{a_y} + \frac{\partial f}{\partial z}\vec{a_z}$$

$$= (2xy)\vec{a_x} + (x^2)\vec{a_y} + (e^z)\vec{a_z}$$

$$= (2(1)(2))\vec{a_x} + (1)^2\vec{a_y} + e^{-1}\vec{a_z}$$

$$\nabla f = 4\vec{a_x} + \vec{a_y} + e^{-1}\vec{a_z}$$

② Find the gradient of given scalar $F = e^{-z}\sin 2x \cosh y$

soln:-

$$\nabla f = \frac{\partial(e^{-z}\sin 2x\cosh y)}{\partial x}\vec{a_x} + \frac{\partial(e^{-z}\sin 2x\cosh y)}{\partial y} + \frac{\partial(e^{-z}\sin 2x\cosh y)}{\partial z}$$

$$= \vec{a_x}\, 2e^{-z}\cos 2x\cosh y + e^{-z}\sin 2x \sinh y\,\vec{a_y}$$

$$+ (-1)e^{-z}\sin 2x\cosh y \cdot \vec{a_z}$$

(3.) $V = \rho^2 z \cos 2\phi$

(4.) $F = 5r \sin^2\theta \cos\phi.$

(3.) soln:-

$$\nabla V = \frac{1}{\rho}\frac{\partial}{\partial\phi}\left(\rho^2 z\cos 2\phi\right) + \frac{\partial}{\partial\rho}\left(\rho^2 z\cos 2\phi\right) + \frac{\partial}{\partial z}\left(\rho^2 z\cos 2\phi\right)$$

$$= -2\rho z\sin 2\phi \, \vec{a_\phi} + 2\rho z\cos 2\phi \, \vec{a_\rho} + \rho^2\cos 2\phi . \vec{a_z}$$

(4.) soln:-

$$\nabla F = \frac{\partial}{\partial r}\left(5r\sin^2\theta\cos\phi\right)\vec{a_r} + \frac{1}{r}\frac{\partial}{\partial\theta}\left(5r\sin^2\theta\cos\phi\right)\vec{a_\theta} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\left(5r\sin^2\theta\cos\phi\right)\vec{a_\phi}$$

$$= 5\sin^2\theta\cos\phi\,\vec{a_r} + 2(5)\sin\theta\cos\theta\cos\phi\,\vec{a_\theta} + 5r\sin^2\theta\sin\phi . \vec{a_\phi}$$

$$= 5\sin^2\theta\cos\phi\,\vec{a_r} + 10\sin\theta\cos\theta\cos\phi\,\vec{a_\theta} + 5\sin^2\theta\sin\phi\,\vec{a_\phi}$$

Divergence of vector:

Divergence measures how much the field diverges or emerges from the point.

The divergence of vector A and point p is defined as the outward flux per unit volume

$$\operatorname{div}\vec{A} = \nabla\cdot\vec{A} = \lim_{\Delta v\to 0}\frac{\oint A . ds}{\Delta v}$$

02/11/26 Types of Divergence:

i) Positive Divergence:

iii) Zero divergence:

ii) Negative divergence:

Divergence in cartesian coordinates/rectangular:

Expression:

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Divergence in cylindrical coordinates:

$$\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial (\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

Divergence in spherical coordinates:

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r\sin\theta} \frac{\partial}{\partial \theta} (A_\theta \sin\theta) + \frac{1}{r\sin\theta} \frac{\partial A_\phi}{\partial \phi}$$

Properties of Divergence:

i) $\nabla \cdot \vec{A} = 0$
The vector field having its divergence zero

ii) $\nabla(\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$

iii) $\nabla(\nabla A) = \nabla^2 A$

iv) $\nabla \cdot (\vec{A}\vec{B}) = \vec{A}(\nabla \vec{B}) + \vec{B}(\nabla \vec{A})$

① Find the divergence of the given vector.

$$\vec{r} = 5x\hat{i} + 2y\hat{j} - 3z\hat{k}$$

soln:-

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot \vec{r} = \frac{\partial r_x}{\partial x} + \frac{\partial r_y}{\partial y} + \frac{\partial r_z}{\partial z}$$

$$= \frac{\partial}{\partial x}(5x) + \frac{\partial}{\partial y}(2y) - \frac{\partial}{\partial z}(3z)$$

$$= 5+2-3$$

$$\nabla \cdot \vec{r} = 4$$

② Evaluate the divergence of a given vector at

$$v = xy\vec{i} + 2yz\vec{j} + 3zx\vec{k} \quad (-1,-1,2)$$

soln:-

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(3zx)$$

$$= y + 2z + 3x$$

$$= (-1) + 2(2) + 3(-1)$$

$$= -1 + 4 - 3$$

$$\nabla \cdot \vec{v} = 0$$

solenoidal.

③ Determine the divergence of the given vector

$$v = x^2 yz\vec{a_x} + xz\vec{a_z}$$

④

$$Q = \frac{\rho \sin\phi}{\partial \rho}\vec{a_\rho} + \frac{\rho^2 z}{\partial \phi}\vec{a_\phi} + \frac{z\cos\phi}{\partial z}\vec{a_z}$$

(5) Find the divergence of vector $\vec{F} = \frac{1}{r^2}\cos\theta\,\vec{a_r} + r\sin\theta\cos\phi\,\vec{a_\theta} + \cos\theta\,\vec{a_\phi}$

(3) soln:-

$$\nabla\cdot\vec{V} = \frac{\partial}{\partial x}(x^2yz) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(xz)$$

$$= 2xyz + x$$

(4) soln:-

$$\nabla\cdot\vec{Q} = \frac{\partial}{\partial\rho}\left(\frac{\rho\sin\phi}{2\rho}\right) + \frac{\partial}{\partial\phi}\left(\frac{\rho^2 z}{2\phi}\right) + \frac{\partial}{\partial z}\left(\frac{z\cos\phi}{2z}\right)$$

$$= \frac{2\rho(\sin\phi) - 2(\rho\sin\phi)}{(2\rho)^2} + \frac{(2\phi)(0) - (\rho^2 z)(2)}{(2\phi)^2} + \frac{(2z)(\cos\phi) - [z\cos\phi](2)}{(2z)^2}$$

(4) soln:-

$$\nabla\cdot\vec{Q} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho Q_\rho) + \frac{1}{\rho}\frac{\partial Q_\phi}{\partial\phi} + \frac{\partial Q_z}{\partial z}$$

$$\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\cdot\frac{\rho\sin\phi}{2\rho}\right) = \frac{1}{\rho}\frac{\sin\phi}{2}$$

$$\frac{1}{\rho}\frac{\partial}{\partial\phi}\left(\frac{\rho^2 z}{2\phi}\right) = \frac{-\cancel{2}(\rho^{\cancel{2}} z)}{\cancel{4}\phi^2}\cdot\frac{1}{\cancel{\rho}} = \frac{-\rho z}{2\phi^2}$$

$$\frac{\partial}{\partial z}\left(\frac{\cancel{2}\cos\phi}{\cancel{2}z}\right) = -\frac{\cos\phi}{z^2}$$

$$\nabla\cdot\vec{Q} = \frac{\sin\phi}{2\rho} - \frac{\rho z}{2\phi^2} - \frac{\cos\phi}{z^2}$$

(5) soln:-

$$\nabla\cdot\vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(\cancel{r^2}\frac{1}{\cancel{r^2}}\cos\theta\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\overset{(r\sin^2\theta\cos\phi)}{\left(r\sin\theta\cos\phi\cdot\sin\theta\right)} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}(\cos\theta)$$

$$= \frac{1}{r^2}(0) + \frac{1}{r\sin\theta}(\cos\theta\cos\phi\, 2\sin\theta) + \frac{1}{r\sin\theta}(0)$$

$$\nabla\cdot\vec{F} = 2\cos\theta\cos\phi$$

Curl of a vector:

Rotation of a vector field around a closed path is known as curl of a vector.

$$\vec{B} = \nabla \times \vec{A} = \lim_{\Delta S \to 0} \frac{\oint A \cdot dl}{\Delta S} \vec{a_n}$$

$\oint_c \vec{A} \cdot dl \Rightarrow$ Circulation of $\vec{A}$ around a closed path.

It can be defined as a maximum circulation of a vector per unit area as $\Delta S \to 0$ whose direction is normal to the surface

Eg:

When a leaf floats in sea water and its rotation is about z axis, velocity is z-direction.

If the curl of the vector is zero, the vector is irrotational $(\nabla \times \vec{A}) = 0$

Curl in cartesian coordinates system: (x, y, z)

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$$

$$\vec{A} = A_x \vec{a_x} + A_y \vec{a_y} + A_z \vec{a_z}$$

$$\nabla \times \vec{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\vec{a_x} - \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z}\right]\vec{a_y} + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\vec{a_z}$$

Curl in cylindrical coordinates: (ρ, ϕ, z)

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \vec{a_\rho} & \rho\vec{a_\phi} & \vec{a_z} \\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

Curl in spherical coordinates:

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \vec{a_r} & r\vec{a_\theta} & r\sin\theta\, a_\phi \\ \partial/\partial r & \partial/\partial\theta & \partial/\partial\phi \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

Properties of curl:

i) The divergence of curl of vector is zero (Rotational operation)

$$\nabla \cdot (\nabla \times \vec{A}) = 0$$

ii) The curl of gradiant of scalar is zero (null properties)

$$\nabla \times (\nabla V) = 0$$

iii) Irrotational vector:

$\nabla \times \vec{A} = 0 \Rightarrow$ curl free field is also called as irrotational vector.

iv) Vector identity:

$$\nabla \times \nabla \times \vec{H} = \nabla(\nabla \cdot \vec{H}) - \nabla^2 H$$

$$\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$$

① Find the curl of a given vector $\vec{A} = x^2yz\,\vec{a_x} + xz\,\vec{a_z}$

② Determine the curl of a given vector $\vec{F} = 30\vec{a_x} + 2xy\vec{a_y} + 5xz^2\vec{a_z}$ $(1, 1, -0.2)$

③ Determine the curl of a given vector $\vec{A} = \rho\sin\phi\,\vec{a_\rho} + \rho^2 z\,\vec{a_\phi} + z\cos\phi\,\vec{a_z}$

④ $\vec{F} = (x + 2y + az)\vec{a_x} + (bx - 3y - z)\vec{a_y} + (4x + cy + 2z)\vec{a_z}$
is irrotational find a, b, c.

$$\nabla \times \vec{F} = 0$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (x+2y+az) & (bx-3y-z) & (4x+cy+2z) \end{vmatrix}$$

$$= \vec{a_x}\left[\frac{\partial}{\partial y}(4x+cy+2z) - \frac{\partial}{\partial z}(bx-3y-z)\right] - \vec{a_y}\left[\frac{\partial}{\partial x}(4x+cy+2z) - \frac{\partial}{\partial z}(x+2y+az)\right] + \vec{a_z}\left[\frac{\partial}{\partial x}(bx-3y-z) - \partial/\partial y\,(x+2y+az)\right]$$

$$= \vec{a_x}\left[c - (-1)\right] - \vec{a_y}\left[4 - a\right] + \vec{a_z}\left(b - 2\right]$$

$$\nabla \times \vec{F} = \vec{a_x}(c+1) + \vec{a_y}(a-4) + \vec{a_z}(b-2) = 0$$

$\vec{a_x}(c+1) = 0$ $\qquad$ $\vec{a_y}(a-4) = 0$

$\boxed{c = -1}$ $\qquad$ $\boxed{a = 4}$

$$\vec{a_z}\,[b-2] = 0$$

$$\boxed{b = 2}$$

① Soln:-

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2yz & 0 & xz \end{vmatrix}$$

$$\nabla \times \vec{A} = \vec{a_x}\left(\frac{\partial(xz)}{\partial y} - \frac{\partial(0)}{\partial z}\right) - \vec{a_y}\left(\frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial z}(x^2yz)\right) + \vec{a_z}\left(\frac{\partial}{\partial x}(0) - \frac{\partial(x^2yz)}{\partial y}\right)$$

$$= \vec{a_x}(0) - \vec{a_y}(z - x^2y) + \vec{a_z}(0 - x^2z)$$

$$= (-z - x^2y)\vec{a_y} + \vec{a_z}(-x^2z)$$

(2) soln:-

(1, 1, -0.2)

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 30 & 2xy & 5xz^2 \end{vmatrix}$$

$$= \vec{a_x}\left(\frac{\partial}{\partial y}(5xz^2) - \frac{\partial}{\partial z}(2xy)\right) - \vec{a_y}\left(\frac{\partial}{\partial x}(5xz^2) - \frac{\partial}{\partial z}(30)\right) + \vec{a_z}\left(\frac{\partial}{\partial x}(2xy) - \frac{\partial}{\partial y}(30)\right)$$

$$= \vec{a_x}(0) - \vec{a_y}(5z^2) + \vec{a_z}(2y)$$

$$= -5z^2\vec{a_y} + 2y\vec{a_z}$$

$$= -5(-0.2)^2\vec{a_y} + 2(1)\vec{a_z}$$

$$\nabla \times \vec{F} = -0.2\vec{a_y} + 2\vec{a_z} \neq 0$$

$\vec{F}$ is irrotational

(5) soln:-

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \vec{a_\rho} & \rho\vec{a_\phi} & \vec{a_z} \\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z \\ \rho\sin\phi & \rho(\rho^2 z) & z\cos\phi \end{vmatrix}$$

$$= \frac{1}{\rho}\left[-z\sin\phi - \rho^3\right]\vec{a_x} - \frac{1}{\rho}\rho\left[0\right]\cancel{\vec{a_y}} + \vec{a_z}\frac{1}{\rho}\left[3\rho^2 z - \rho\cos\phi\right]$$

$$= \vec{a_\rho}\left(\frac{-z\sin\phi}{\rho} - \rho^2\right) + (3\rho z - \cos\phi)\vec{a_z}$$

5/1/26

Divergence theorem:

Total out~~put~~ward flux per unit volume.

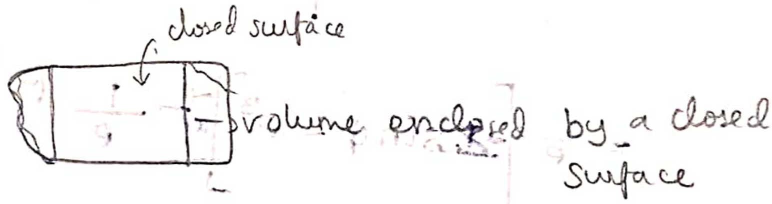
Consider a volume with the out~~put~~ward flux.

Statement:

The total outward flux of a vector field to the closed surface S is equal to the volume integral of divergence of vector

$$\oiint_S \vec{A} \cdot ds = \iiint_V \nabla \cdot \vec{A} \, dv$$

It converts the surface integral of normal component of any vector into a volume integral of the divergence of the vector.



[flux $\perp^r$ direction]

Proof:

$$\vec{A} = A_x \vec{a_x} + A_y \vec{a_y} + A_z \vec{a_z}$$

RHS:

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{A_x} + \frac{\partial A_y}{A_y} + \frac{\partial A_z}{A_z}$$

Take volume integral on both sides

$$\Rightarrow \iiint_V \nabla \cdot \vec{A} \, dv = \iiint_V \left[\frac{\partial A_x}{A_x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right] dx dy dz \rightarrow (1)$$

$$\iiint_V \frac{\partial A_x}{\partial x}\, dx\,dy\,dz + \iiint_V \frac{\partial A_y}{\partial y}\, dx\,dy\,dz + \iiint_V \frac{\partial A_z}{\partial z}\, dx\,dy\,dz$$

$$\Rightarrow \iint_V \left[\int \frac{\partial A_x}{\partial x}\, dx\,dy\,dz\right] = \iint_V \left[\int (\partial A_x)\, dy\,dz\right]$$

$$= \iint A_x\, dy\,dz$$

similarly

$$= \iint A_x\, ds_x \qquad ds_x = dy\,dz$$

$$\Rightarrow \iiint_V \frac{\partial A_y}{\partial y}\, dx\,dy\,dz = \iint A_y\, ds_y$$

$$\Rightarrow \iiint \frac{\partial A_z}{\partial z}\, dx\,dy\,dz = \iint A_z\, ds_z$$

sub all values in (1)

$$\iiint_V \nabla\cdot\vec{A}\, dv = \iint A_x\, ds_x + \iint A_y\, ds_y + \iint A_z\, ds_z$$

$$= \iint \left[A_x\, ds_x + A_y\, ds_y + A_z\, ds_z\right]$$

$$\because \left[\begin{array}{l}\vec{A} = A_x + A_y + A_z \\ ds_x + ds_y + ds_z = ds\end{array}\right]$$

$ds \rightarrow$ surface area of a perpendicular direction

$$\iiint_V \nabla\cdot\vec{A}\, dv = \oiint \vec{A}\cdot ds$$

Hence proved.

Applications:

* It converts volume integral into surface integral and vice versa.
* It applies to both time varying and static field
* It is widely used in the field of Engineering, Dynamic etc...

Problems in divergence theorem:

(1) State and verify the divergence theorem for the vector $\vec{A} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ taken over the bounded by $x=0, x=1, y=0, y=1$ & $z=0, z=1$

Soln:

Statement:

$$\iiint_V \nabla \cdot \vec{A}\, dv = \iint \vec{A}\, ds$$

LHS:

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$= \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2)$$

$$\nabla \cdot \vec{A} = 4 - 4y + 2z$$

$$\iiint_V \nabla \cdot \vec{A}\, dv = \int_0^1\int_0^1\int_0^1 (4-4y+2z)\, dx\, dy\, dz$$

$$= \int_0^1\int_0^1 \left[4x - 4xy + 2xz\right]_0^1 dy\, dz$$

$$= \int_0^1\int_0^1 \left[4 - 4y + 2z\right] dy\, dz$$

$$= \int_0^1 \left[4y - \frac{4y^2}{2} + 2yz\right]_0^1 dz$$

$$= \int_0^1 \left[4 - 2 + 2z\right] dz$$

$$= \left[4z - 2z + \frac{2z^2}{2}\right]_0^1$$

$$= 4 - 2 + 1$$

$$\boxed{\iiint_V \nabla \cdot \vec{A}\, dv = 3} \rightarrow (1)$$

RHS:

$\iint$

1/1/26 R.H.S:

$$\oiint \vec{A}\cdot ds = \iint A\cdot\vec{i}\,dydz + \iint A\cdot(-\vec{i})\,dydz + \iint A\cdot\vec{j}\,dxdz$$
$$\iint A\cdot(-\vec{j})\,dxdz + \iint A\cdot\vec{k}\,dxdy + \iint A\cdot(-\vec{k})\,dxdy$$

$$\iint A\cdot\vec{i}\,dydz = \int_0^1\int_0^1 (4x\vec{i} - 2y^2\vec{j} + z^2\vec{k})\,\vec{i}\,dydz$$
$$= \int_0^1\int_0^1 4x\,dydz$$
$$= \int_0^1 [4xy]_0^1\,dz$$
$$= \int_0^1 4x\,dz$$
$$= [4xz]_0^1$$
$$\iint A\cdot\vec{i}\,dydz = 4x$$

$\because x=1$ $\quad \iint A\cdot\vec{i}\,dydz = 4$

$\because x=0$ $\quad \iint A\cdot(-\vec{i})\,dydz = 0 \Rightarrow$ for $4x(\vec{i})\cdot -\vec{i} = -4x$

Check the validity of the divergence theorem

$$\iint A\cdot\vec{j}\,dxdz = \int_0^1\int_0^1 (4x\vec{i} - 2y^2\vec{j} + z^2\vec{k})\,\vec{j}\,dxdz$$
$$= -\int_0^1\int_0^1 2y^2\,dxdz$$
$$= -\int_0^1 [2xy^2]_0^1\,dz$$
$$= -\int_0^1 2y^2\,dz$$
$$= -[2y^2z]_0^1$$
$$\iint A\cdot\vec{j}\,dxdz = -2y^2$$

$\because x=1 \quad = -2$

$$\iint A(-\vec{j})\,dxdz = +\int_0^1\int_0^1 2y^2\,dxdz$$
$$= 2y^2$$

$\because x=0 \quad = 0$

$$\iint A\vec{k}\,dx\,dy = \int_0^1\int_0^1 (4x\vec{i}-2y^2\vec{j}+z^2\vec{k})\,\vec{k}\,dx\,dy$$

$$= \int_0^1\int_0^1 z^2\,dx\,dy$$

$$= \int_0^1 [xz^2]_0^1\,dy$$

$$= \int_0^1 z^2\,dy$$

$$= [yz^2]_0^1$$

$$= z^2$$

$\because x=1 \quad = 1$

$$\iint A(-\vec{k})\,dx\,dy = -z^2$$

$\because x=0 \quad = 0$

$$\oiint \vec{A}\cdot ds = 4+0-2+0+1+0$$

$$\oiint \vec{A}\cdot ds = 3 \rightarrow (2)$$

$$(1) = (2)$$

Hence divergence theorem is verified.

Stoke's theorem:

Stoke's theorem relate the line integral and surface integral

$$\oint \vec{H} \cdot d l = \iint_{S} (\nabla \times \vec{H}) \, ds$$

~~Proof~~ The line integral of the vector around a closed path is equal to the surface integral of the normal component of its curl over any closed surface.

Proof:

The surface S is subdivided into K in number of cells.

Let K cell has surface area ∇Sk bounded by dk

For total closed path:

$$\oint_{L} \vec{A} \cdot d\vec{l} = \sum_{k} \oint_{k} \vec{A} \cdot dl$$

Circulation of $\vec{A}$ around lk

multiply and divide by ΔSk

$$\oint_{L} \vec{A} \cdot d\vec{l} = \sum_{k} \left(\frac{\oint_{Lk} \vec{A} \, dl}{\Delta Sk} \right) \Delta Sk$$

$$= \sum_{k} (\nabla \times \vec{A}) \nabla Sk$$

where ΔSk is the smaller surface area

$$\oint \vec{A} \cdot d\vec{l} = \iint \nabla \times \vec{A} \, ds$$

$$\oint_{k} \vec{A} \cdot d\vec{l} = \iint \nabla \times \vec{A}$$

Uses:

It is mainly used to convert the surface integral into line integral and vice versa.

To find line integral of some particular curve

To determine the curl of the bounded surface.

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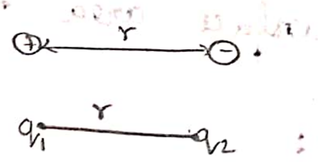
Coulomb's law:

Coulomb's law is mainly used to analyse electrically charge particles



$$F \propto \frac{q_1 q_2}{r^2}$$

Distance r increases, Force decreases

$$F \propto \frac{1}{r^2}$$

The force of attraction or repulsion between any two point charges is directly proportional to the product of two charges.

$$F = \frac{q_1 q_2}{4\pi\varepsilon r^2} \text{ N}$$

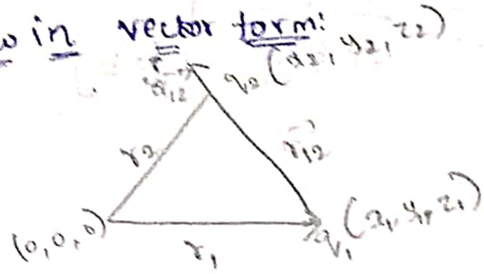
$\frac{1}{4\pi\varepsilon} \Rightarrow$ proportional constant

$\epsilon = \epsilon_0 \epsilon_r$

$\epsilon_0 = 8.85 \times 10^{-12}$ F/m

ϵ_r = relative permittivity = 1

Coulomb's law in vector form:



$$\vec{F} = \frac{q_1 q_2}{4\pi\epsilon \vec{r_{12}}^2} \vec{a_{12}}$$

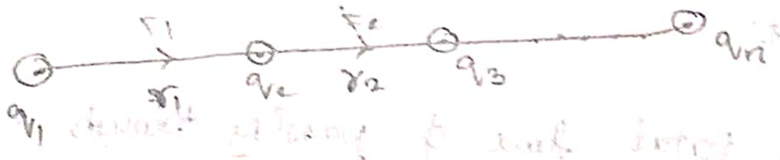
Coulomb's law: characteristics of vector force:

* It has direction & magnitude
* $\vec{a_{12}}$ ⇒ Unit direction of r_{12}

$$\vec{a_{12}} = \vec{r_2} - \vec{r_1} \Big/ |r_1 - r_2|$$

$$r_{12} = r_2 - r_1$$

Principle of superposition:



Consider a system of end point of charges $q_1, q_2 \dots q_n$ with distance $r_1, r_2 \dots r_{n-1}$. Force between charges $F_1, F_2, \dots F_n$.

Superposition principle:

$$\vec{F_i} = \sum_{j=1}^{n} \frac{q_i q_j}{4\pi\epsilon r_{ij}^2} \times \vec{a_{ij}} \quad \Big| \; i \neq j$$

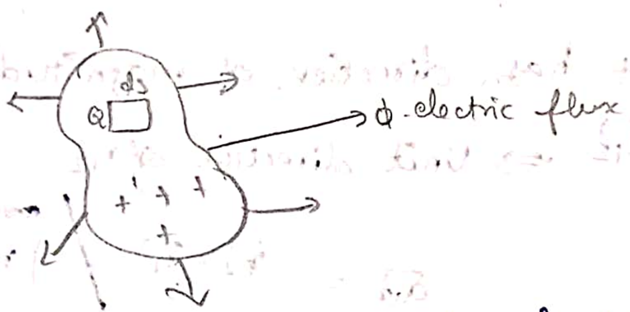
The force on n^{th} charge is given by vector sum of all the individual forces

~~superposition principle~~

The force F_i on the i^{th} charge is given as $\bar{F_i} = \sum_{j=1}^{n} \frac{q_i q_j}{4\pi \varepsilon r_{ij}^2} \Big| i \neq j$

Gauss's law:

Gauss's law relates total flux linked with a closed surface and total charge enclosed by the surface.



It relates the density of the electric flux

$D = \frac{\phi}{A}$

$\phi = DA$

ϕds

$\phi = \oint D \cdot ds$

Statement:

The total flux ϕ passing through a closed surface is $\frac{1}{\varepsilon_0}$ times the charge enclosed in the surface.

$\phi = \oint D\, ds = Q$

$\oint \varepsilon_0 E\, ds = Q$

$\oint E\, ds = \frac{1}{\varepsilon_0} Q$

$D \propto E$

$D = \varepsilon_0 E$

$\varepsilon_r = 1$

ε_0 = constant

Gauss's law in point form:

$$\oint D \cdot ds = Q \rightarrow (1) \rightarrow \text{integral}$$

According to gauss's law

If the volume charge density is ρv then the net charge enclosed by the volume is

$$Q = \int_v \rho v \, dv \rightarrow (2)$$

sub (2) in (1)

$$\oint_s D ds = \int_v \rho v dv \rightarrow (3)$$

By applying Divergence theorem is LHS

$$\oint D \cdot ds = \int_v \nabla \cdot D dv \rightarrow (4)$$

compare (3) & (4)

$$\int_v \nabla \cdot D dV = \int_v \rho v \cdot dv$$

$$\nabla \cdot D = \rho v \rightarrow \text{point form}$$

$$\nabla \cdot \varepsilon_0 E = \rho v$$

$$\nabla \cdot E = \frac{\rho v}{\varepsilon_0}$$

It ~~states~~ states that the divergence of electric flux density is equal to the volume charge density

Gaussian surface:

The surface over which gauss's law is applied is called as Gaussian surface.

Conditions:

Many condition the surface is closed

The electric flux density D is either normal or tangential to the surface at each point.

The electric flux density D is constant over the part of the surface where D is normal.

Applications:

* Used to determine charge Q or electric field intensity for symmetrical charge distribution.

Eg:
Infinite line charge
Single cell of charge
Coaxial cylinders

Electrostatics:

It is also called static electric field. It is study of charge at rest condition

$\vec{E}$ - Electric field

Characteristics of $\vec{E}$:

* Electric field lines ~~or~~ electric flux $\to \psi$
* Electric field lines ~~or~~ electric flux density, D
* Electric field lines ~~or~~ electric field intensity — E

• Electric field lines ~~or~~ electric potential - V

Electric field lines:

i) Electric field lines are imaginary lines which give visual idea of electric field.

ii) It can never be a closed loop.

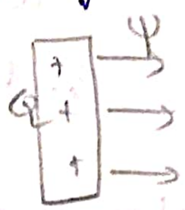
iii) It can never cross each other.

Electric flux:

It is the total number of electric lines of force producing from the charge.

If Q is the charge of body then the electric flux produced from that is given by

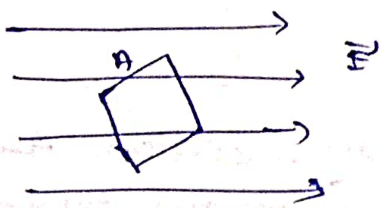
$\psi = Q$.



Electric flux density:

It is defined as the amount of flux passing through the unit surface area perpendicular to the direction of the flux.

Unit : C/m^2



Electric potential:

$$V \propto q/r$$

$$V = \frac{q}{4\pi \varepsilon_0 r}$$